



Associate in Applied Science **Simulation and Game Development** (A25450B) Programming Concentration

Overview

Turn your passion for games into a career. Central Piedmont is home to the first state-approved game-development degree in the country, and this program immerses you in the full craft—design, programming, and 3D modeling and animation—using the same industry-standard tools studios rely on, from Unreal Engine and Unity to Maya, ZBrush, and the Adobe suite.

The Programming concentration develops the code behind the play—game programming, database and AI systems, mobile development, and software engineering—so you can build the systems games run on.

Employment Outlook

The game industry is booming, and the skills you'll build reach well beyond it. Graduates work in entertainment, education, healthcare, engineering, and simulation—anywhere interactive 3D and software come together—and hands-on, portfolio-driven coursework plus work-based learning get you ready for real studio work.

Your degree can open doors to roles like Game Programmer, Gameplay Engineer, Tools Programmer, Mobile Game Developer, and AI Programmer.

(Please note: CPCC does not guarantee employment)

Plan of Study

CREDITS

Term I

SGD 111 Introduction to Simulation and Game Development	3
SGD 112 SGD Design I	3
SGD 113 SGD Programming I	3
SGD 114 SGD 3D Modeling I	3
SGD 212 Simulation and Game Development Design II	3

You may have completed program certificate C25450-21. Confirm eligibility with your academic advisor.

Term Credits 15

Term II

Humanities/Fine Arts Elective	3
ENG 111 Writing and Inquiry	3
ACA 122 Transfer & Career Success	1
Social/Behavioral Sciences Elective	3
MAT 121 Algebra/Trigonometry I	3
COM 231 Public Speaking	3

Term Credits 16

Term III

SGD 165 SGD Character Development	3
SGD 213 Simulation Game Development Programming II	3

Term Credits 6

Term IV

SGD 122 Simulation and Game Database Programming	3
SGD 125 Simulation and Game Artificial Intelligence	3
SGD 289 Simulation and Game Development Capstone Project	3
SGD 285 SGD Software Engineering	3
SGD 214 SGD 3D Modeling II	3
WBL 111 Work-Based Learning I	1

Term Credits 16

Term V

SGD 168 SGD Mobile Programming I	3
SGD 172 SGD Virtual Environments	3
WBL 121 Work-Based Learning II	1
SGD 162 SGD 3D Animation I	3
SGD 268 SGD Mobile Programming II	3

You may have completed a program certificate(s). Confirm eligibility with your academic advisor.

Term Credits 13

Total Credit Hours

66